Project Title
NEWS EXPLORER

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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Submitted by
G V MADHAV 2520030568
K RICHIE KIRAN 2520030498
BOJJA VISHNU VARDHAN 2520030392

Team Number: 5
Section: 6

Under the guidance of

Dr. V. Sireesha

Professor, Department of CSE

Hyderabad-500090, Telangana, India.

ABSTRACT

News Explorer is an advanced DSA based news analysis system designed to demonstrate the practical application of advanced algorithms to real-world text data. The system processes and explores news articles using efficient string-processing, dynamic programming, graph, and probabilistic techniques. It supports functionalities such as fast pattern searching using KMP or Rabin-Karp, fuzzy text matching through edit distance, document similarity, and analysis of relationships between news content. Advanced concepts such as suffix arrays, LCP, network flow, approximation algorithms, and randomized algorithms are explored where applicable. The project emphasizes implementing core algorithms manually rather than relying on built-in Java utilities, helping demonstrate their underlying logic and efficiency. By combining multiple advanced algorithmic techniques in a single practical application, News Explorer provides an interactive platform for understanding how theoretical concepts from advanced data structures and algorithms can solve real-world information retrieval and text-analysis problems.

Signature of the Guide